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SUPERSEDING
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FEDERAL SPECIFICATION

INK, DRAWING, BLACK AND COLORS: (FOR GLAZED PLASTIC SURFACES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for drawing inks suitable for use on glazed plastic surfaces.

1.2 Classification.

1.2.1 *Types.* The drawing inks shall be of the following types, as specified (see 6.2):

Type I.—Transparent permanent.
Type II.—Opaque permanent. (See 3.8.1.)
Type III.—Opaque water soluble.

1.2.2 *Classes (type II only).* Type II drawing ink shall be of the following classes, as specified (see 6.2).

Class 1.—Medium consistency.
Class 2.—Light consistency.
Class 3.—Heavy consistency.

1.2.3 *Colors.*

1.2.3.1 *Type I.* Type I drawing ink shall be blue, brown, green, orange, purple, red, or yellow, as specified (see 6.2).

1.2.3.2 *Types II and III.* Unless otherwise specified, types II and III drawing inks shall be black (see 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-390—Plastic, Molding Material, Polyethylene, Low and Medium Density.
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-636—Boxes, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 — Boxes, Set-Up, Paperboard.
PPP-T-45—Tape, Gummed, Paper, Reinforced.
PPP-T-97 — Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Hand-

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books and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.* Type I ink shall be composed of a transparent solution of a dye in a liquid medium (see 4.4.1).

3.1.2 *Types II and III.* Types II and III inks shall be a suspension of an opaque pigment in a liquid medium. Dye and pigment may be combined if desired (see 4.4.1).

3.2 Drawing quality. When tested as described in 4.4.2, the ink shall flow freely and uniformily without dropping from the pen and shall not gum, dry, or flake on the pen while being used. The ink shall adhere perfectly to cellulose acetate. The drawn lines shall be clean, unbroken, of uniform width, and free from bleeding or feathering. The intersecting lines shall be free from distortion, or blots caused by the ink spreading under the ruling edge.

3.3 Drying quality. Lines drawn with the ink and allowed to dry for 10 minutes shall not blur, smudge, or chip off when rubbed with the fingers (see 4.4.2).

3.4 Permanency.

3.4.1 *Fading.* When tested as described in 4.4.3.1, the ink shall show no evidence of fading.

3.4.2 *Water fastness.* When tested as described in 4.4.3.2, types I and II drawing inks when dry shall be insoluble in water.

3.5 Water solubility. Type III ink shall be soluble in water and completely removed from the plastic sheet when tested as described in 4.4.3.3.

3.6 Stability. There shall be no mold, scum, deposit, or sediment when tested as described in 4.4.4.

3.7 Thinner. The contractor shall have available for purchase a suitable ink thinner for use with the type I and type II drawing inks he proposes to furnish under this specification. When specified, the thinner, in the quantity specified, shall be furnished with the ink.

3.8 Containers. Ink shall be packaged in containers having a minimum of 3/4, 2, 8, and 16 ounces, as specified (see 6.2). Three-fourth-ounce containers shall have a quill, or equivalent, extending to within 1/8 inch of the bottom of container, but shall not touch the bottom.

3.8.1 *Sets (type II only).* Type II ink shall also be available in sets containing the following:

Item	*Quantity*
Class I ink—2-oz. bottle	3
Class II ink—2-oz. bottle	3
Class III ink—2-oz. bottle	4
Thinner—2-oz. bottle	2

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his

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own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.2.)

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* All bottles of drawing ink of the same type, class, and color offered for delivery at one time shall be considered a lot for inspection.

4.2.2 *Sampling for examination.* A random sample of drawing ink shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.2.3 *Sampling for tests.* A random sample of drawing ink shall be selected from each lot in accordance with MIL-STD-105, at inspection level S-1.

4.3 Examination. Each filled bottle selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table I. Any bottle of drawing ink in the sample containing one or more defects shall be rejected and if the number of defective bottles of drawing ink in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE I.—*Classification of defects*

Categories	Defects
Critical:	None defined.
Major:	
101	Not type, class, or color specified.
102	Bottle closure not type specified.
103	Bottle leaks.
104	Contents less than specified.
105	Marking not as specified or illegible. (see 5.3.)
Minor:	None defined.

4.3.1 *Examination of preparation for delivery.* An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5 of this specification. Defects shall be scored in accordance with table II. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery, selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-4 with an AQL of 4.0 percent defective.

TABLE II.

Examine	Defects
Markings (Unit, intermediate and exterior containers, as applicable)	Incorrect; omitted; illegible; improper size, location, sequence, or method of application.
Container materials	Wrong type, class, color, or other category of required specifications.
	Material missing, damaged, or defective affecting serviceability.
Workmanship	Wrong or careless means in placement and use of labeling and packaging and packing materials such as loose or partially attached labels, incompatible adhesives, flaps of containers incompletely closed, loose or overtight straps, dangerous strap ends, protruding nails, and smeared adhesive.
Contents	Quantities of unit and intermediate containers not as specified.
	Wrong arrangement; bottle on side or upside down.
Gross weight of shipping container	In excess of specified weight.
	In excess of box limitation shown on boxmaker's certificate.

4.4 Tests. Drawing ink selected in accordance with 4.2.3 shall be tested as specified herein. Drawing ink failing any test will be considered cause for rejection of the lot represented by the sample.

4.4.1 *Determination of transparency and opacity.* Draw lines with a broad-stroke

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lettering pen, 1/8-inch wide with black, waterproof, acetate drawing ink on clear-finish cellulose acetate sheet, 0.005 inch thick and allow to dry thoroughly. With a sable or ox hair show card brush, paint strips 1/2-inch wide with each color over the black lines. If the black lines show clearly through the colors, the inks shall be classified as transparent dye inks (type I). If the colors substantially obliterate the black lines, the inks shall be classified as opaque pigment inks (types II and III).

4.4.2 *Drawing (type I; type II, class* 1; *and type III).* A properly sharpened detail pen shall be used. It should always be charged with, as nearly as possible, the same volume of ink, and there shall be no change in the setting of the pen at any time during the progress of this test. The ruler used shall have a square edge, that is, it shall not be beveled. Set the pen with an opening of approximately 0.025 inch and draw 10 parallel lines 6 inches long, 1/4 inch apart on a clear-finish cellulose acetate sheet 0.005 inch thick. After these lines are allowed to dry for 10 minutes, 10 parallel lines of the same width and length shall be drawn at a right angle across the lines previously drawn. Examine the drawn lines for conformance to 3.2. Conformance to 3.3 shall also be determined by rubbing the drawn lines gently with dry finger tips 10 minutes after the lines have been drawn.

4.4.2.1 *Type II, classes* 2 *and* 3. The above test shall be performed except for the pen. For class 2, a fine lettering pen shall be used. For class 3, a ruling pen shall be used.

4.4.3 *Water and light fastness.* Cut the cellulose acetate sheet upon which the test in 4.4.2 has been made into three strips with approximately the same number of drawn lines on each strip. One strip each shall be used for the tests in 4.4.3.1, 4.4.3.2 (types I and II) and 4.4.3.3 (type III). The third strip shall be reserved for comparison after these tests. Keep the strips away from light and fumes.

4.4.3.1 *Light fastness.* One strip selected in 4.4.3 shall be exposed to direct sunlight for 24 hours, or at a distance of about 10 inches from a glass-enclosed carbon arc lamp such as the Fade-Ometer, type FDA-R or its equivalent, for 12 hours. Comparison shall be made with the reserved strip.

4.4.3.2 *Water fastness* (types I and II). Allow strip selected in 4.4.3 to dry for at least 24 hours. Drop 1 ounce of tap water on strip and spread it over the drawn lines, using no rubbing or friction. Allow the water to dry naturally. The lines shall show no evidence of running, bleeding, smearing, or bleaching. Comparison shall be made with the reserved strip.

4.4.3.3 *Water solubility.* Allow the strip selected in 4.4.3 to dry for at least 1 hour. Rub the drawn lines with a wet cloth. Excessive rubbing should not be required to remove ink.

4.4.4 *Stability.* Place 100 milliliters of the sample ink in a container, cover the container with watchglass to keep out dust and lint, yet permitting access of air, and let ink remain in container for 15 days. After this period of time, examine for evidence of mold, scum, deposit or sediment.

4.4.5 *Permeability test.* Five ink-filled 3/4-ounce polyethylene containers of each inspection sample and five ink-filled 3/4-ounce glass containers of the same ink representing control samples shall be sealed with paraffin wax along the edge of the stoppers, and individually weighed at room temperature. The ink and container shall then be placed inside a temperature-controlled test chamber and maintained at a temperature of 100° ± 1°F. for 28 consecutive days. After this period of time, the containers shall be removed, allowed to cool at room temperature for 1 day, and reweighed. The average weight loss of the test samples minus the average weight loss of the control samples shall be the average weight loss of the test samples due to permeation.

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If the average weight loss is greater than 0.4 gram per 3/4-ounce container, the lot shall be rejected. (see 5.1.2.1.)

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing specified shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Ink shall be furnished in commercial type bottles. Each bottle shall be packaged in a folding carton or set-up box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.1.1 *Intermediate packaging.* Twelve 3/4-ounce bottles shall be further packaged, in folding carton or set-up boxes, conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Two-ounce, eight-ounce, and sixteen-ounce bottles in quantities as specified (see 6.2), and individual sets of type II ink shall be further packaged in fiberboard boxes conforming to PPP-B-636. Box closures shall conform to the appendix to the applicable box specification.

5.1.2 *Level B.*

5.1.2.1 *Unit package.* Three-quarters, two, eight, and sixteen ounces of ink shall be packaged in 3/4-, 2-, 8-, and 16-ounce bottles, respectively, and shall be furnished as specified (see 6.2). The bottles shall be fabricated from commercial grade glass or from polyethylene material conforming to type I, grade 1 of L-P-390. If a polyethylene container is used, the average weight loss due to permeation of the ink through the container shall not exceed 0.4 gram per 3/4-ounce container when tested as specified in 4.4.5. The shape of the 3/4-ounce container shall be such that it can be inserted in the ink bottle holders commercially available. Each 3/4-ounce bottle shall be provided with a leakproof stopper and ink applicator in accordance with standard commercial practice. The stopper and applicator shall not be affected by the ink, nor shall the ink be affected by the stopper and applicator. The applicator shall extend down to within 1/8 inch of the bottom of the bottle when the stopper is tightly in place. Each 2-, 8-, and 16-ounce bottle shall be provided with a plastic continuous-thread screw cap closure (see 5.1.2.1.1).

Caps shall be of adequate weight sidewall so that an average of not more than 10 out of 30 caps will be broken in application by capping machines at 30 inch-pounds torque and no individual cap will break at less than 15 inch-pounds. All screw caps shall have a nonabsorbent resilient lining or a lining backed by resilient material which will remain in the cap. An inner seal shall be used to cover and adhere to the mouth of the 2-, 8-, and 16-ounce bottles which will prevent loss of the contents should the cap become loosened. Caps shall be tightened to prevent leakage. Each glass bottle and each 3/4-ounce plastic bottle shall be packaged in a snug-fitting box conforming to style II, type A, class a of PPP-B-566. Boxes containing bottles of 2-ounce size shall have tuck ends of a lock type as illustrated in figure 2 of PPP-B-566 for the style of box specified. In addition, each box containing 16-ounce glass bottles shall be fitted with a single-faced corrugated liner. Each individual box containing 8- or 16-ounce bottles shall be sealed with tape as specified in the appendix to the box specification.

5.1.2.1.1 *Alternate closure.* Alternate non-metal continuous-thread screw-cap closure for the 2-, 8-, and 16-ounce size bottles, of pourout-type or other design in accordance with normal commercial practice, will be acceptable provided the alternate closure meets the quality established by this specification and is at least equally effective to that described above for the purpose intended.

5.1.2.2 *Intermediate package.* Ink, unit packaged as specified in 5.1.2.1, except for

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16-ounce size bottles, shall be intermediate-packaged in quantities as specified in table III. Sixteen-ounce bottles shall be intermediate-packaged in boxes conforming to type I or II, class 1 or 2, W6c or W6s of PPP-B-636 or PPP-B-665; in quantities as specified in table III.

TABLE III.

Size (ounces)	Quantity of unit packages per intermediate package	Quantity of unit packages per shipping container
3/4	[1]12	144
2	[1]12	144
8	6	96
16	6	48

[1] *Alternate intermediate packaging*: Two boxes, each tightly wrapped with kraft paper and each containing 6 unit packages as specified in 5.1.2.1, may be inserted in a snug-fitting box (intermediate package) as specified in 5.1.2.2 to comply with the requirement for 12 unit packages per intermediate package.

5.1.2.2.1 *Sets (type II only)*. Each individual set (see 3.8.1) shall be intermediate-packaged in a close-fitting box conforming to PPP-B-636, type I or II, class 1, style RSC. Interior slotted partitions shall be used to form cells for each bottle. The partitions shall be made of the same type material as used in the box construction. Each box shall be closed in accordance with the appendix to the box specification.

5.1.3 *Level C*. Ink, in bottles as specified herein, shall be packaged in accordance with normal commercial practice.

5.2 Packing. Ink, packaged as specified in 5.1, shall be packed level A, B, or C, as specified (see 6.2).

5.2.1 *Level A*. Ink shall be packed in fiberboard boxes conforming to PPP-B-636, class 2, V3c. Gross weight shall not exceed the weight limitations of the box specification. Box closures and strapping shall conform to the appendix to the box specification. In lieu of steel strapping, boxes may be reinforced with pressure-sensitive filament tape conforming to PPP-T-97, type IV, applied in accordance with the appendix thereto. Boxes shall be provided with caseliners conforming to MIL-L-10547, and sealed in accordance with the appendix thereto.

5.2.2 *Level B*. Bottles, packaged as specified in 5.1, shall be packed in quantities as specified in table III in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636, except that the minimum bursting strength of the fiberboard shall be 200 p.s.i. and each box shall be furnished with top and bottom pads of minimum 200 p.s.i. test, double-faced "C" flute fiberboard. All flaps of fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed, by application throughout the entire area of contact between the flaps; by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive; or by using tape not less than 3 inches in width conforming to PPP-T-45, applied over center seams and extending over ends for a distance of not less than 2-1/2 inches.

5.2.2.1 *Sets (type II only)*. Not more than 24 sets, packaged as specified in 5.1.2.2.1, shall be packed in shipping containers as specified in 5.2.2.

5.2.3 *Level C*. Unless otherwise specified, individual bottles and sets packaged as specified in 5.1, shall be packed in shipping containers of the type, size, and kind commonly used in commercial practice, in a manner that will insure acceptance by common carriers and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies*. In addition to any special marking specified in the contract or order, intermediate packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123, including specification classification (such as type, class, color, etc.). Bottles and unit packages

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shall be marked in accordance with the manufacturer's normal commercial practice and, except for type II ink in sets, shall include the Federal Stock Number.

5.3.2 *Military activities.* In addition to any special marking specified in the contract or order, intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bottles and unit packages shall be marked in accordance with the manufacturer's normal commercial practice and, except for type II ink in sets, shall include the Federal Stock Number.

6. NOTES

6.1 Intended use. Drawing inks covered by this specification are intended for use on plastic surfaces, such as, celluloid, cellulose acetate, and vinylite and are applied by pen or brush. Type II drawing inks are used as follows:

Class 1—medium—for general art and drafting use.
Class 2—light—for fine lettering pen use.
Class 3—heavy—for ruling pen.

6.2 Ordering data. Purchasers should exercise any desired options offered herein (see 4.1 and 5.2.3) and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and color required (see 1.2.1, 1.2.2, and 1.2.3).
(*c*) Size of container required, other than in sets (see 3.8 and 3.8.1).
(*d*) Both the level of packaging and the level of packing required and any special marking for shipment (see 5.1, 5.2, and 5.3).

6.3 Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

MILITARY CUSTODIANS:

Army—Quartermaster Corps

Navy—Bureau of Supplies and Accounts

Air Force—Mobile Air Materiel Area

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